

PROFILE

I build and verify compiler and runtime components. My main project is UniversalToolchain/Wist2, a modular .NET SDK for language composition with typed contracts, deterministic planning, interpreter/CIL execution, and callable-aware SSA. I am seeking junior/intern roles in compiler, runtime, developer tools, and language-platform engineering.

KEY EVIDENCE

1,579 tests · 0 failures

Public verification manifest dated August 2, 2026: 1,579 tests, zero failures; nine packages and clean-consumer scenarios.

1st of 49

1st among 49 submitted solutions; the only 5,0/5,0 score and 104/104 official tests in the LLVM compiler track selection.

Correctness verification

Exact-oracle, differential, and metamorphic testing, replay/reduction, and conservative fail-closed decisions.

EXPERIENCE

- MCST — Compiler Engineering Intern

LLVM · C++23 · remote

July — August 2026

- Developing an LLVM optimization pass.
 - Implemented cycle-detecting RPO, Dijkstra, Dinic maximum flow, and Tarjan SCC over a shared C++23 graph model; CMake build with input/output checks and a Python test runner.

[GitHub](#)
- UniversalToolchain/Wist2 — Creator and Primary Developer

.NET SDK · language platforms · runtime

2024 — present

- Designed language composition from independently versioned packages: typed artifacts, capabilities/conflicts, deterministic routes, pass ordering, and exact manifest/lock binding.
 - Implemented interpreter/CIL execution, callable-aware SSA, and verifier-gated transformations; unresolved configuration and semantic states fail closed.
 - Public verification manifest dated August 2, 2026: 1,579 tests, zero failures; nine packages and clean-consumer scenarios.

[GitHubDocumentationVerification](#)

EXPERIENCE — CONTINUED

Programming Instructor and Technical Mentor2021 — present

Algorithms · languages · execution models

- Teach algorithms, C/C++, C#, Python, ABI fundamentals, and execution models; review code and design learning paths.

SELECTED PROJECTS

PS-form Memory Dependence Analyzer

C17 · program analysis

- C17 analyzer for parametric memory-access dependence with conservative yes/no/maybe outcomes.
- Symbolic normalization, residue/GCD abstractions, exact affine analysis, and cost-guarded fallbacks; a harness with an exact oracle, metamorphic WA hunting, and model-guided TL hunting.

GitHub

x86-64 Codegen & Register Allocation Playground

Rust · x86-64 · register allocation

- Experimental playground for x86-64 code generation, linear-scan register allocation, and reproducible correctness/performance checks.
- SSA-like IR, liveness, assignment verification, and differential parity with an IR interpreter; explicitly not positioned as a production-ready backend.

GitHub

SKILLS

Compilers and runtime	Bytecode/AIR, CFG, dominance, SSA, optimizations, interpreter/CIL, LLVM, program analysis
Languages	C#/.NET, C++23, C17, Python, Rust
Verification	exact oracles, differential/metamorphic testing, replay, reduction, CI, sanitizers
Tools	Linux, Git, GitHub Actions, CMake, ASP.NET Core, PostgreSQL/SQLite

EDUCATION AND RECOGNITION

HSE University — Software Engineering program, since 2026.	HSE Vysshaya Proba prize-winner in competitive and industrial programming; two-time first-degree diploma recipient at the MEPhI Junior competition.
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Wist evidence source: [f13ad1310856](#) · canonical 642ccd8810cd · 2026-08-03.1